Program : **Diploma in Robotic Process Automation**	
Course Code : **5308**	Course Title: **Data Mining Lab**
Semester: **5**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- Learn to perform data mining tasks using a data mining toolkit (such as open source WEKA).
- Understand the data sets and data preprocessing.
- Demonstrate the working of algorithms for data mining tasks such association rule mining, classification, clustering and regression.
- Exercise the data mining techniques with varied input values for different parameters.
- To obtain Practical Experience Working with all real data sets.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic principles and theorems of Engineering Mathematics	1002 2002	Mathematics I &II	1 & 2
Basic knowledge of Database	3133	Database Management System	3

Course Outcomes:

On completion of the course student will be able to:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Use different features of WEKA tool	6	Applying
CO2	Preprocess the data for mining	9	Applying

CO3	Determine association rules	12	Applying
CO4	Model various classifiers & Examine clusters from the available data	15	Applying
	Series Test	3	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	2	2	3	3		
CO2	3	2	2	3	3		
CO3	3	2	2	3	3		
CO4	3	2	2	3	3		
CO5	3	2	2	3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Use different features of WEKA tool**		
M1.01	Basics of WEKA tool a. Investigate the Application interfaces. b. Explore the default datasets	3	Applying
M1.02	Creating a new Arff File	3	Applying
CO2	**Preprocess the data for mining**		
M2.01	Pre-process a given dataset based on Attribute Selection	3	Applying
M2.02	Pre-process a given dataset based on Handling missing values	3	Applying

M2.03	Pre-process a given dataset based on Discretization	3	Applying
	Series Test – I	1.5	
CO3	**Determine association rules**		
M3.01	Calculate information Gain	3	Applying
M3.02	Build a Decision Tree using ID3 algorithm.	3	Applying
M3.03	Generate Association Rules using the FP-Growth algorithm.	3	Applying
M3.04	Accuracy and error Measures evaluation	3	Applying
CO4	**Model various classifiers & Examine clusters from the available data**		
M4.01	Demonstrate classification process on a given dataset using Naïve Bayesian Classifier.	3	Applying
M4.02	Demonstrate classification process on a given dataset using Nearest neighbor Classifier.	3	Applying
M4.01	Build a distance matrix of the given data using various distance measures.	3	Applying
M4.02	Cluster the given dataset by using the k-Means algorithm and visualize the cluster mean values and standard deviation of dataset attributes.	3	Applying
M4.03	Cluster the given dataset using a hierarchical clustering algorithm.	3	Applying
	Series Test – II	1.5	

**** - Micro Project**

Students are expected to do a micro project in data mining during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record.
Example: Develop program such as

- **Disease Prediction**
- **Housing Price Prediction**
- **Fake News Detection**

Text / Reference

T/R	Book Title/Author
T1	Dunham M H, "Data Mining: Introductory and Advanced Topics", Pearson Education, New Delhi, 2003.
T2	Jaiwei Han and Micheline Kamber, "Data Mining Concepts and Techniques", Elsevier, 2006.

Online Resources

Sl.No	Website Link
1	https://www.investopedia.com/terms/d/datamining.asp
2	https://www.techtarget.com/
3	https://www.javatpoint.com/data-mining
4	https://www.electronicshub.org/artificial-neural-networks-ann/
5	https://www.tutorialspoint.com/data_mining/
6	https://towardsdatascience.com/the-fp-growth-algorithm

List of Experiments:

1. Basics of WEKA tool a. Investigate the Application interfaces. b. Explore the default datasets.

2. Pre-process a given dataset based on the following: a. Attribute Selection b. Handling Missing Values

3. Pre-process a given dataset based on the following: a. Discretization b. Eliminating Outliers

4. Create a dataset in ARFF (Attribute-Relation File Format) for any given dataset and perform Market-Basket Analysis.

5. Generate Association Rules using the FP-Growth algorithm.

6. Build a Decision Tree using ID3 algorithm.

7. Demonstrate classification process on a given dataset using Naïve Bayesian Classifier.

8. Demonstrate classification process on a given dataset using Nearest neighbor Classifier.

9. Build a distance matrix of the given data using various distance measures.

10. Cluster the given dataset by using the k-Means algorithm and visualize the cluster mean values and standard deviation of dataset attributes.

11. Cluster the given dataset using a hierarchical clustering algorithm.